

Selection of Random Effects in Intensive Longitudinal Data Analysis

Lijin Zhang

lijinzhang@stanford.edu

Agenda

01 Introduction

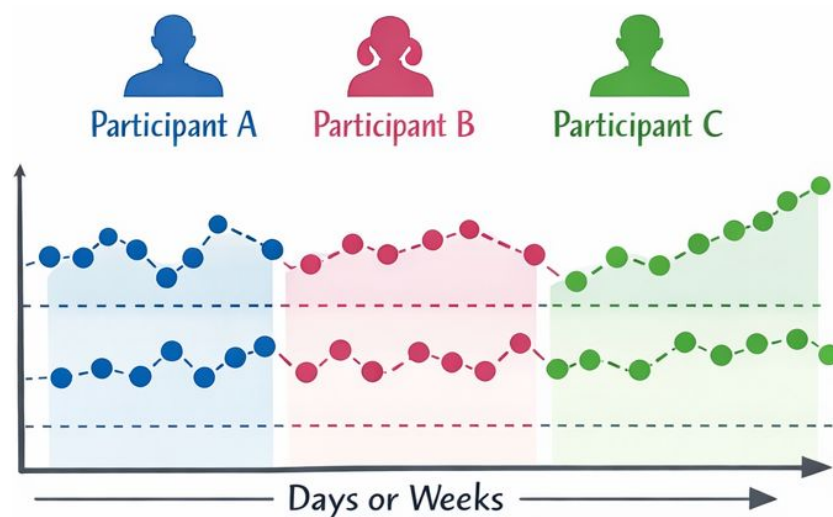
02 The Proposed Method

03 Simulation Studies

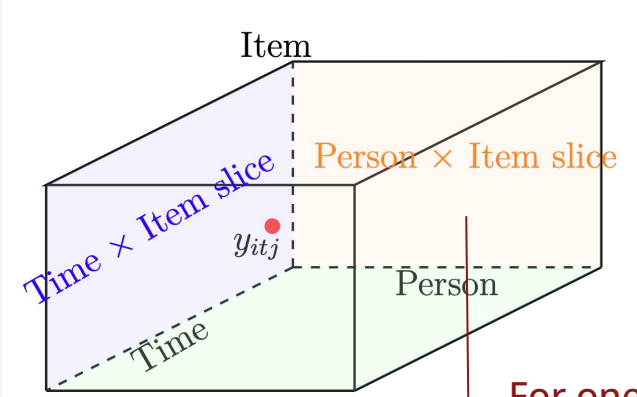
04 Discussion

Intensive Longitudinal Data (ILD)

- Experience Sampling Method/Ecological Momentary Assessment
- Frequent repeated measurements for the same **individuals** over **time**
- **Individual** differences & changes over **time**



Data Structure



For one time point

N\Item	1	2	...	J
1	3	4	...	2
2	1	2	...	3
...		...		
N	4	3	...	1

(a) Person $\times$ Item

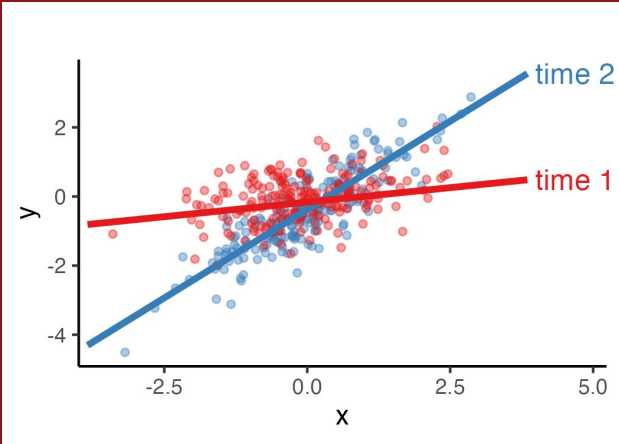
Time 1

N\Item	1	2	...	J
1	3	4		
2	1	2		
...		..		
N	4	3		

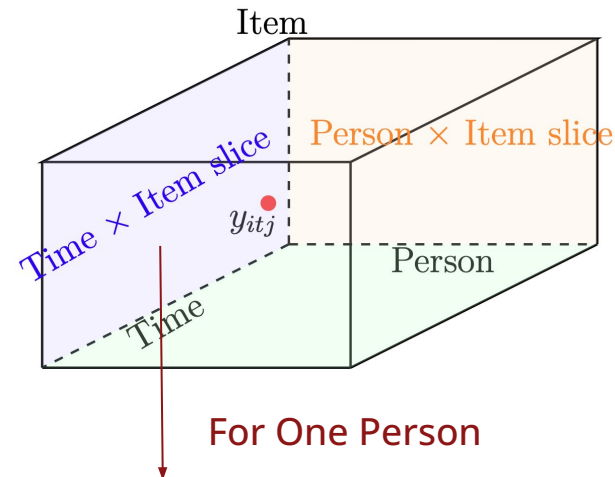
N\Item	1	2	...	J
1	3	4	...	2
2	1	2	...	3
...				
N	4	3	...	1

Time 2

Two Time Points

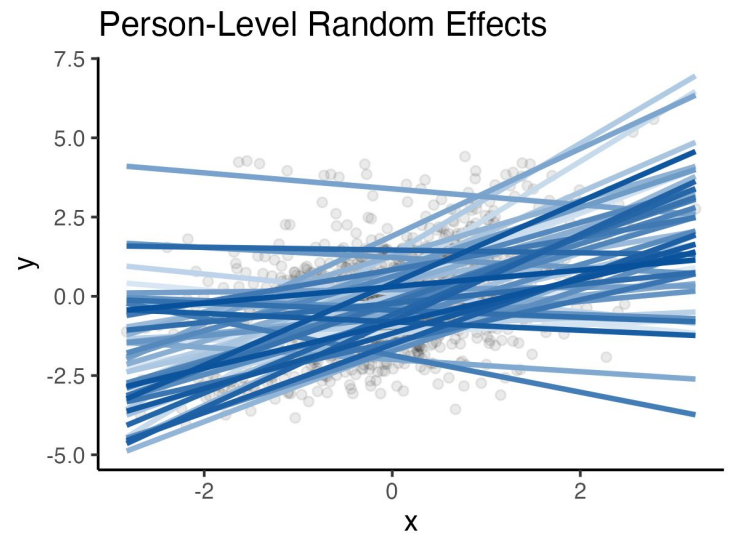


Random Effects across Person



T\Item	1	2	...	J
1	2	1	...	3
2	4	3	...	2
...		...		
T	3	5	...	4

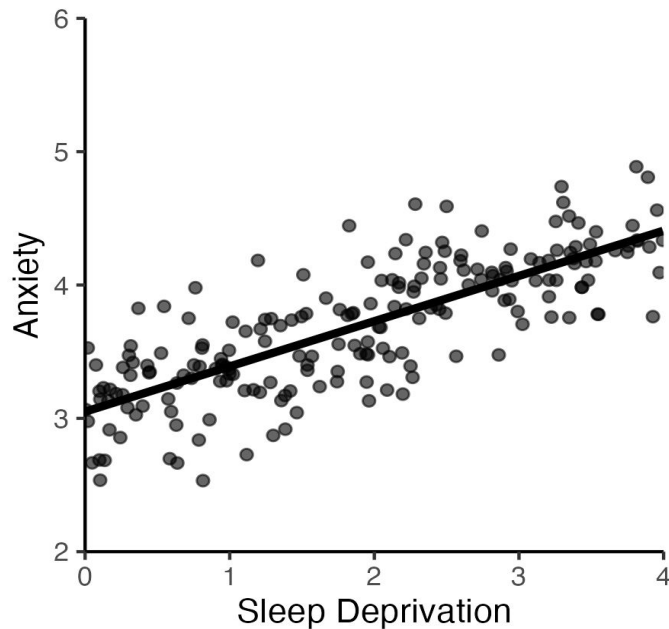
(b) Time $\times$ Item



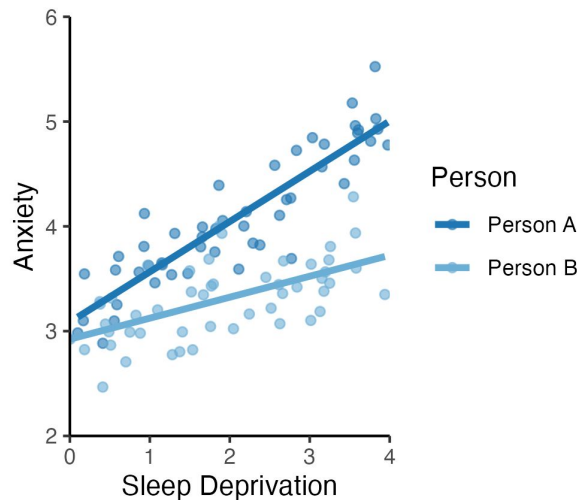
Simulated example: Each line shows the fitted regression for a person.

Person- and Time- level random effects

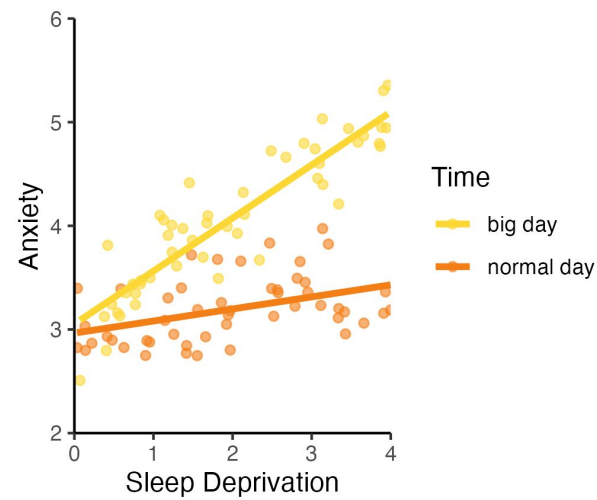
Simulated example for illustration: Differences across person or time



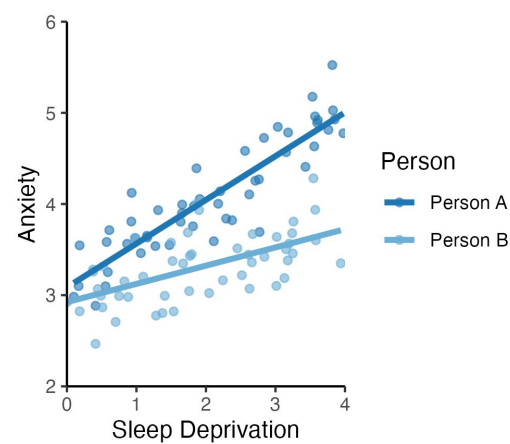
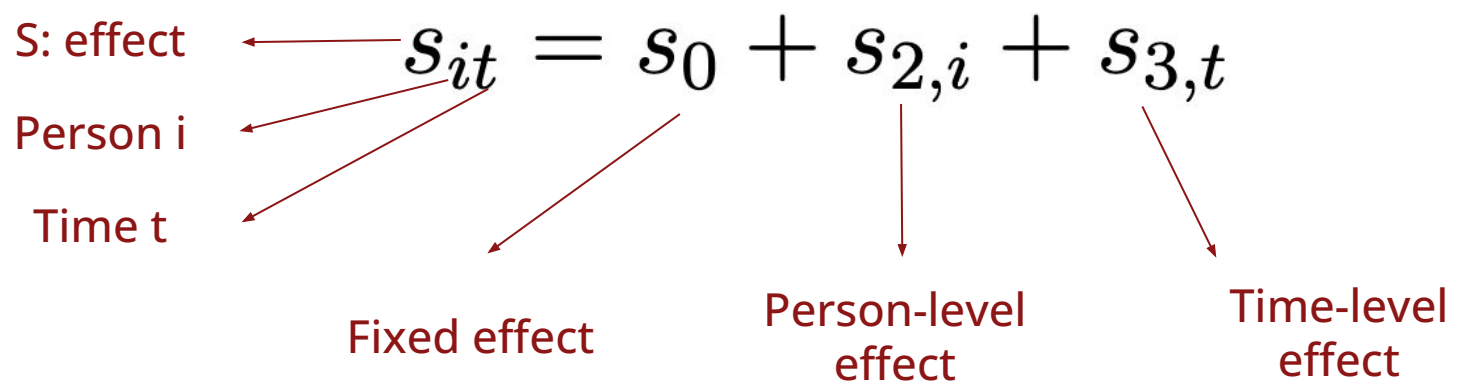
Person level difference



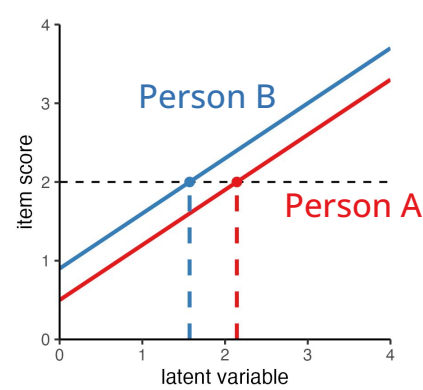
Time level difference



Person- and Time- level random effects



Meaningful
heterogeneity
for investigation



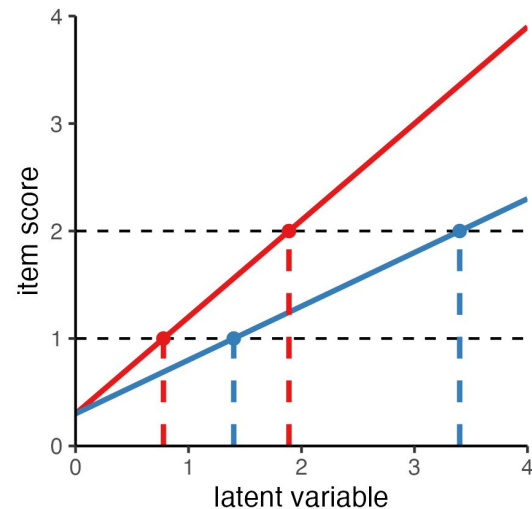
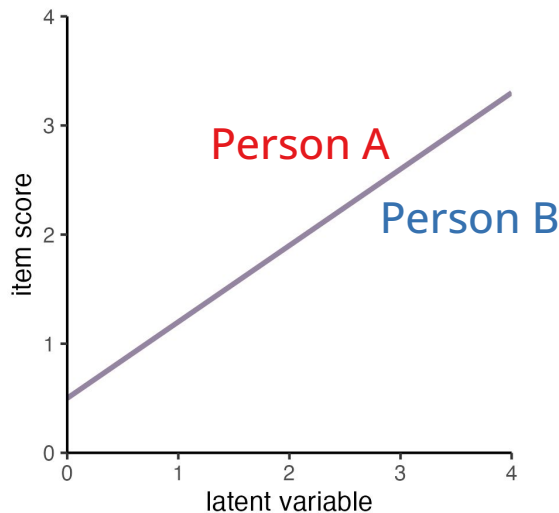
Measurement
non-invariance

Selection of random effects in measurement models

$$s_{it} = s_0 + \boxed{s_{2,i}} + s_{3,t}$$

$s_{2,i} = 0$ for all i

$s_{2,i} \neq 0$ for some i



Traditional Selection

1. **Free:** freely estimate the parameter across people (or time)
2. **Evaluate:** Examine the variance of the parameter (slope or intercept)

$$\text{var}(s_i)$$

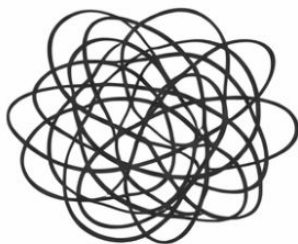
3. **Fix:** Variance below cutoff (e.g., < 0.02 ; Kim et al., 2023) $\rightarrow$ Invariant

$$\text{var}(s_i) < 0.02 \Rightarrow s_{2,i} = 0, \forall i \Rightarrow s_i = S_0$$

Challenges

1. No universal cutoff
2. All measurement / structural effects can be random
3. In measurement model, each item adds 4 potential random effects
 - Person intercept, person slope, time intercept, time slope(5 items $\rightarrow$ 20 effects)

High dimensional model



Slow convergence



Agenda

□ 01 Introduction

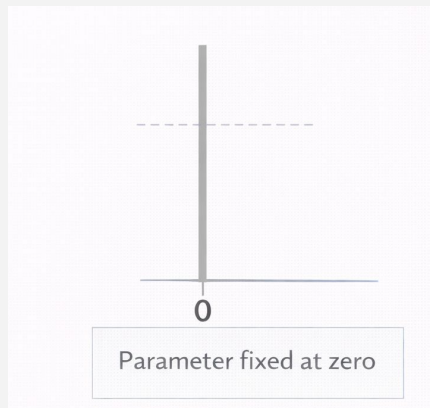
□ **02 The Proposed Method**

□ 03 Simulation Studies

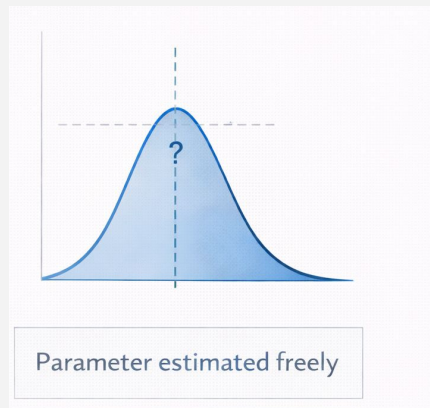
□ 04 Discussion

$$s_{it} = s_0 + \boxed{s_{2,i}} + s_{3,t}$$

When we use variance to decide:

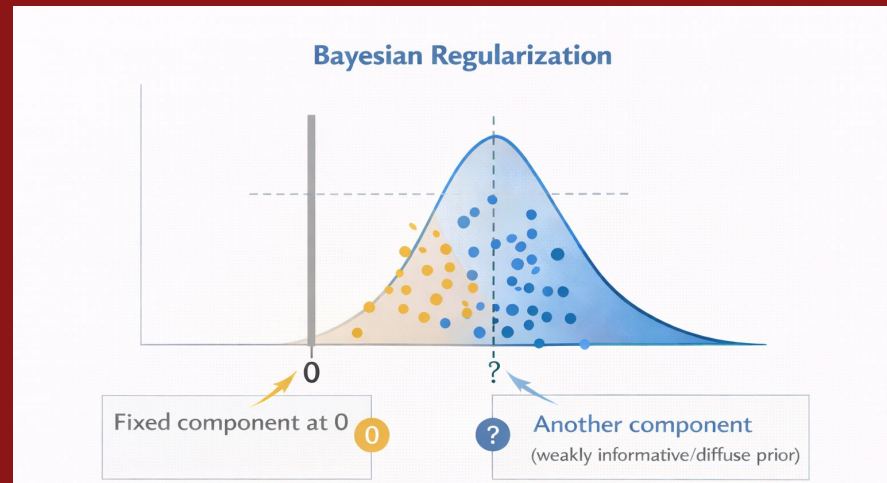


Invariant



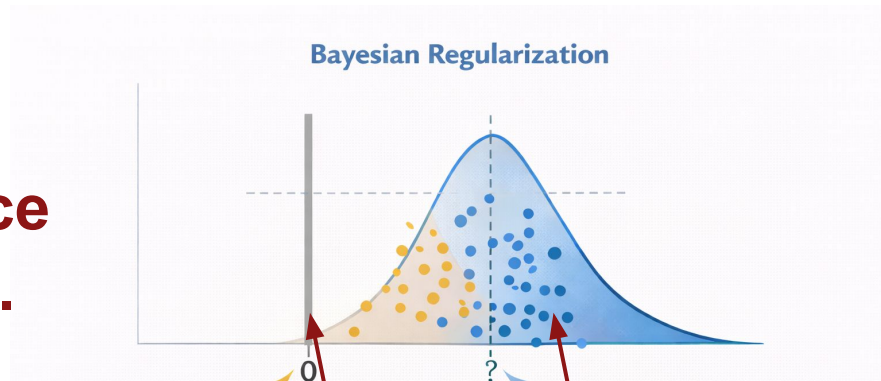
Non-invariant

Chen et al., 2003



Spike-and-Slab Method

**JAGS cannot
assign a variance
of exactly zero.**



Spike $\delta_0 = 0$

Slab $s_{2,i}^* \sim \mathcal{N}(0, \sigma_{s_2}^2)$

$$s_{2,i} = (1 - \pi_{s_2}) \delta_0 + \pi_{s_2} s_{2,i}^*$$

Probability that this
effect is random

$\pi_{s_2} \sim \text{Bernoulli}(\theta_{s_2})$

The Proposed Method

Step 1 — Select

Fit the spike-and-slab model. Retain effects with posterior $\theta > 0.5$.



Step 2 — Re-estimate

Refit with only the selected random effects — removes shrinkage bias.

Where This Applies

Measurement Model

- Factor loadings
- Intercepts
- → *measurement invariance testing*

Structural Model

- Autoregressive & cross-lagged paths
- Time-varying covariate effects
- → *dynamic heterogeneity selection*

Two simulation studies evaluate the method in each setting.

Agenda

01 Introduction

02 The Proposed Method

03 Simulation Studies

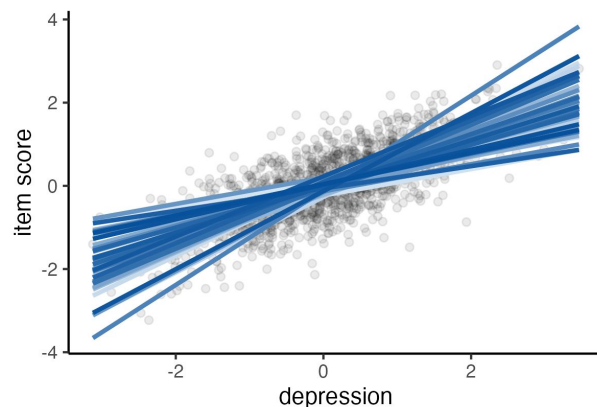
04 Discussion

Simulation Study 1: Measurement Model

We generated Intensive Longitudinal Data for four items (16 potential random effects)

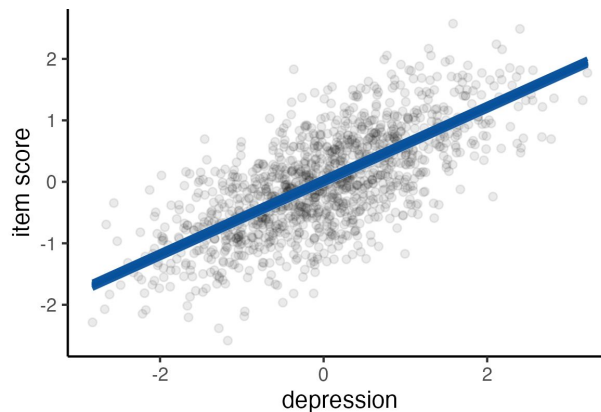
- Proportions of non-invariant effects (50%, 25%, 0%) across both time and person

Non-invariance (0.1)

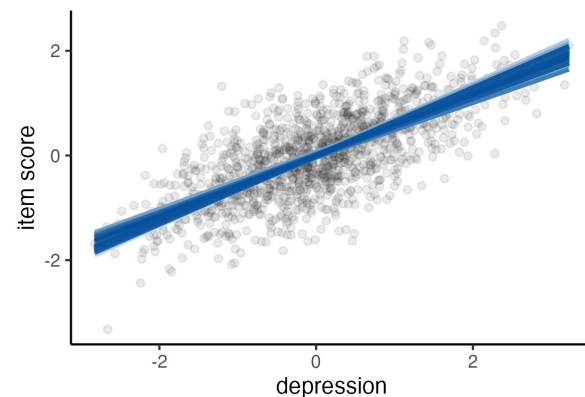


Invariance

Exact Invariance 0



Approximate Invariance 0.001 (Muthén & Asparouhov, 2014)



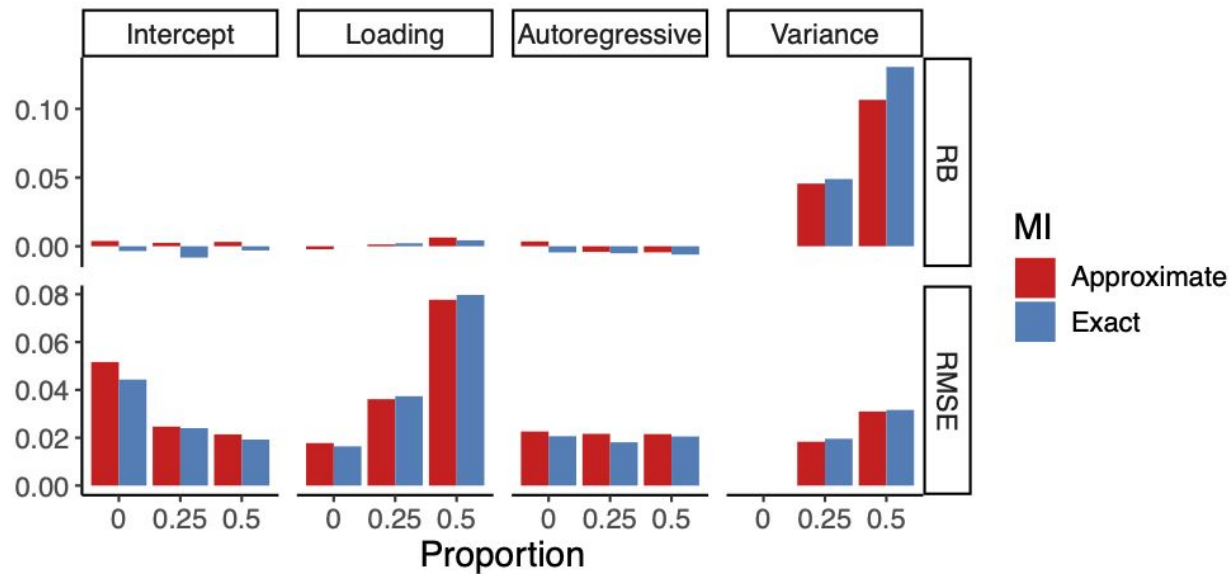
Simulation Results

- High accuracy in identifying measurement non-invariance
- Low error rate

		Incorrectly identify non-invariant		invariant parameter	
		Correctly identify non-invariant	invariant parameter		
MI	Proportion	TPR	TNR	FPR	FNR
Exact	50%	1.000	0.995	0.005	0.000
	25%	1.000	0.995	0.005	0.000
	0%	-	0.997	0.004	-
Approximate	50%	1.000	0.974	0.026	0.000
	25%	1.000	0.929	0.071	0.000
	0%	-	0.932	0.068	-

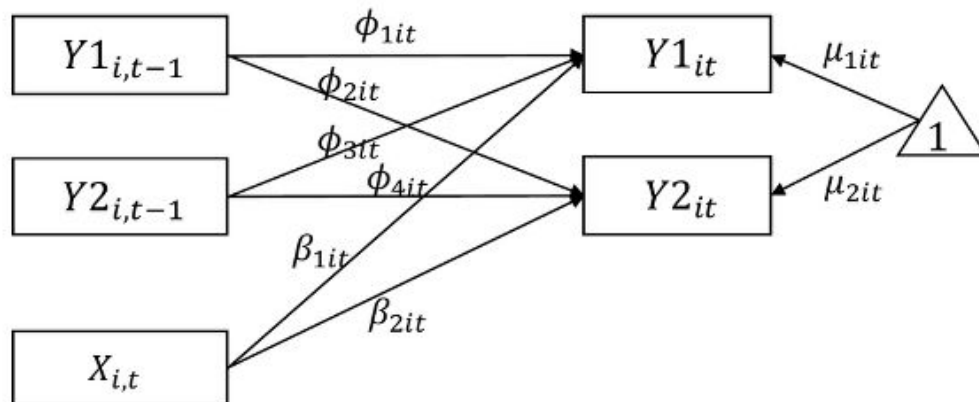
Simulation Results

- Low relative bias in estimation ($<10\%$)
- Low root mean square error



Simulation Study 2: DSEM

- Bivariate dynamic model: Y1, Y2 + time-varying covariate X
- AR paths, cross-lagged paths, covariate effects — each decomposed into person + time deviations
- → 12 candidate random effects
 - (4 dynamic paths + 2 covariate effects) × 2 levels



Simulation Study 2: DSEM

Factor	Levels	Values
Time points (T)	2	25, 50
True random-effect structure	3	0% (none), 50% (6 of 12), 100% (all 12)

- N = 200 persons (fixed)
- **True random-effect variance = 0.02 when present**

Simulation Study 2: DSEM

Table 4.3: Selection Accuracy by Random-Effect Category in DSEM.

T	Proportion	Parameter	TPR	TNR	FPR	FNR
25	100%	Autoregressive	1.000	-	-	0.000
		Cross-lagged	1.000	-	-	0.000
		Covariate	0.942	-	-	0.058
	50%	Autoregressive	0.995	0.969	0.031	0.005
		Cross-lagged	0.995	0.995	0.005	0.005
		Covariate	0.985	0.990	0.010	0.015
	0%	Autoregressive	-	0.998	0.003	-
		Cross-lagged	-	1.000	0.000	-
		Covariate	-	0.998	0.003	-
	100%	Autoregressive	0.997	-	-	0.003
		Cross-lagged	1.000	-	-	0.000
		Covariate	1.000	-	-	0.000
50	50%	Autoregressive	1.000	1.000	0.000	0.000
		Cross-lagged	0.981	1.000	0.000	0.019
		Covariate	1.000	0.994	0.006	0.000
	0%	Autoregressive	-	1.000	0.000	-
		Cross-lagged	-	1.000	0.000	-
		Covariate	-	0.998	0.003	-

Simulation Study 2: DSEM

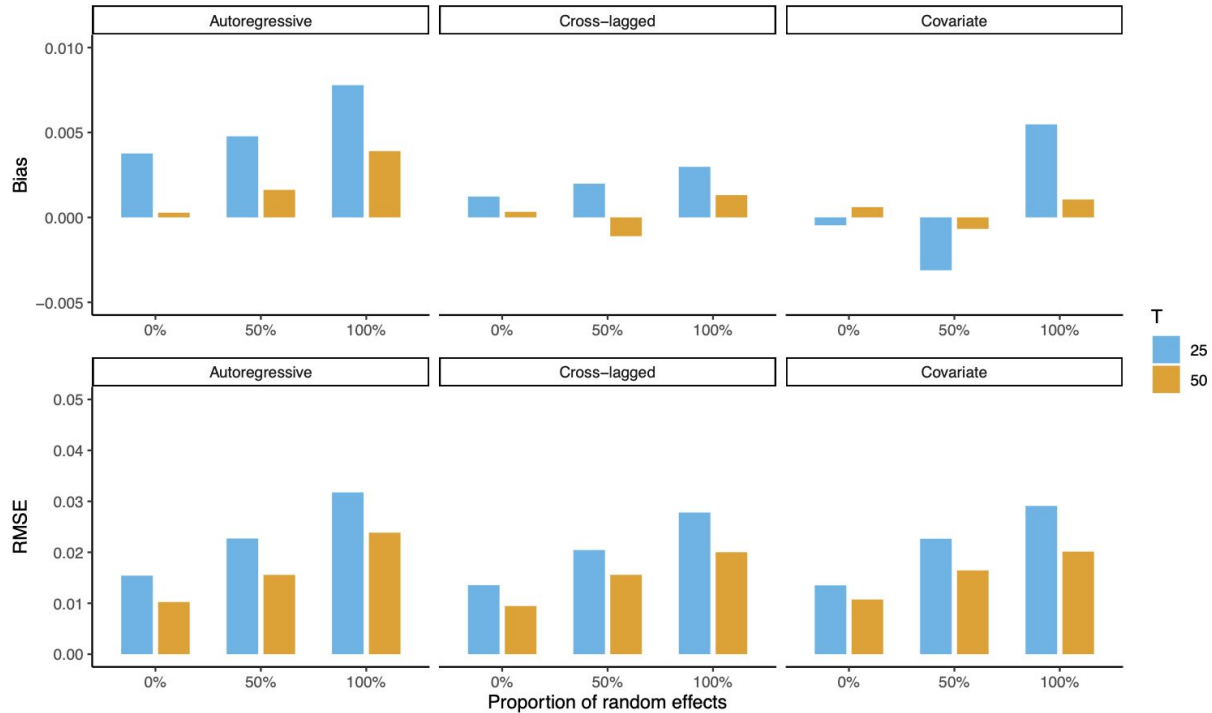


Figure 4.6: Estimation Accuracy in DSEM

Agenda

01 Introduction

02 The Proposed Method

03 Simulation Studies

04 Discussion

Key Takeaways

A spike-and-slab prior replaces fixed variance thresholds with a data-driven decision about which random effects to retain.

The method works simultaneously at the person and time levels, for both measurement (CCFA) and structural (DSEM) parameters.

Selection accuracy stays high across conditions.

In the empirical example, it recovered a substantively meaningful pattern that a fixed 0.02 threshold would have erased.

Thank You!



Ben Domingue



Esther Ulitzsch



Junhao Pan



Yuyin Wang

Email: lijinzhang@stanford.edu

Website: <https://www.lijinzhang.com>

Zhang, L., Ulitzsch, E., Wang, Y., Pan, J., & Domingue, B.W. (manuscript drafted). Selection of Random Effects in Dynamic Structural Equation Modeling